EXP 12

Data Reshaping Using the Air Quality Dataset

Aim:

To perform data reshaping using the built-in `airquality` dataset, including melting, casting, and calculating monthly averages for Ozone, Solar.R, Wind, and Temperature.

Procedure:

1. Load the `airquality` dataset.
2. Remove missing values for the monthly averages.
3. Melt the dataset into a long format.
4. Cast the data into a wide format.
5. Calculate monthly averages for the required variables.

Code:

```r
if (!requireNamespace("reshape2", quietly = TRUE)) {
  install.packages("reshape2", repos = "https://cloud.r-project.org")
}

library(reshape2)

data("airquality")

print(head(airquality))

air_data <- na.omit(airquality)

melted_data <- melt(
  air_data,
  id.vars = c("Month", "Day"),
  measure.vars = c("Ozone", "Solar.R", "Wind", "Temp"),
  variable.name = "Variable",
  value.name = "Value"
)

print(head(melted_data))

cast_data <- dcast(
  melted_data,
  Month + Day ~ Variable,
  value.var = "Value"
)

print(head(cast_data))

monthly_avg <- aggregate(
  cbind(Ozone, Solar.R, Wind, Temp) ~ Month,
  data = air_data,
  FUN = mean
)

print(monthly_avg)
```

OUTPUT:

1. First six rows of the airquality dataset

```
  Ozone Solar.R Wind Temp Month Day
1    41     190  7.4   67     5   1
2    36     118  8.0   72     5   2
3    12     149 12.6   74     5   3
4    18     313 11.5   62     5   4
5    NA      NA 14.3   56     5   5
6    28      NA 14.9   66     5   6
```

2. Melted data output

	Month	Day	Variable	Value
1	5	1	Ozone	41
2	5	2	Ozone	36
3	5	3	Ozone	12
4	5	4	Ozone	18
5	5	7	Ozone	23
6	5	8	Ozone	19

3. Cast data output

	Month	Day	Ozone	Solar.R	Wind	Temp
1	5	1	41	190	7.4	67
2	5	2	36	118	8.0	72
3	5	3	12	149	12.6	74
4	5	4	18	313	11.5	62
5	5	7	23	299	8.6	65
6	5	8	19	99	13.8	59

4. Monthly average output

	Month	Ozone	Solar.R	Wind	Temp
1	5	23.61538	181.29630	11.622222	65.54839
2	6	29.44444	190.16667	10.266667	79.10000
3	7	59.11538	216.42308	8.941667	83.88462
4	8	59.96154	171.85714	8.793103	83.96154
5	9	31.44828	167.43333	10.180000	76.90000